

build-poster-pdf: irregular layouts

Row spans, column spans, and the two ways to write them(build-poster-pdf)

This page is itself an irregular layout. The grid: header below places every box by coordinate, including the tall box on the right and the wide boxes at the bottom.

The two notations

Both `layout:` and `grid:` place boxes on a CSS Grid. They differ only in **how you write the same arrangement**.

- `layout:` – a **matrix of heading names**, written to look like the page.
- `grid:` – an **x/y/w/h coordinate per box**, like a dashboard library.

`grid:` wins if you write both.

The layout matrix

Repeat a name on several rows to span them.

```
layout:
- [INTRO, INTRO, TALL]
- [A, B, TALL]
- [WIDE, WIDE, TALL]
```

`TALL` appears on all three rows, so it becomes one box three rows high. `INTRO` and `WIDE` each appear twice on one row, so they are two columns wide.

The shape of the text is the shape of the page, which is the point of this notation.

The grid coordinates

Give each box a position and a size.

```
grid:
columns: 3
boxes:
- {name: INTRO, x: 0, y: 0, w: 2}
- {name: TALL,  x: 2, y: 0, h: 3}
- {name: A,     x: 0, y: 1}
- {name: B,     x: 1, y: 1}
- {name: WIDE,  x: 0, y: 2, w: 2}
```

x/y count from **0** at the top left. w/h default to **1**. The same arrangement as the matrix on the left, written once per box.

A tall box

This box is h: 3, so it spans **three rows** – it stays beside the three boxes on its left instead of being cut into pieces.

A tall box is the usual home for the one thing that needs the most room: a large results table, an ordination plot, or a long species list.

Because a box grows to fit its contents, a tall box does **not** stretch its neighbours. The row heights come from whatever is tallest in that row.

- h: 2 – spans two rows
- h: 3 – spans three rows (this box)
- w: 2 – spans two columns (see the boxes below)

Which one to use

Neither is better; they trade off against each other.

	layout:	grid:
Reads like the page	yes	no
Spanning box	repeat the name	w: / h: once
Editing one box	retype every affected row	change that box' s numbers
Many irregular boxes	matrix gets noisy	stays compact

A rule of thumb: **start with layout:**, and switch to `grid:` when the repeated names start getting in the way.

What gets checked

A wrong arrangement is caught before the PDF is written, so a broken poster never reaches the printer quietly.

- **Overlap** – two boxes on the same cell are named in the error.
- **Overflow** – a box whose x + w passes columns is named.
- **Missing** – a heading in the body but not in `grid:/layout:` (or the reverse) stops the build.
- **Page count** – the build reports the page count; a poster is always 1, so a 2 means something overflowed the paper.

The last one is a warning rather than an error, because the PDF is still worth looking at to see *what* overflowed.

This very page

The header of this file uses `grid:` with `columns: 3`. The box you are reading is w: 3 (full width), the tall box on the right is h: 3, and the two notation boxes sit side by side in the middle row.

Compare with `poster_howto.md` (a tour of the features), `poster_howto2.md` (input and output side by side), and `golf_course.md` (a realistic poster that uses the `layout: matrix`).